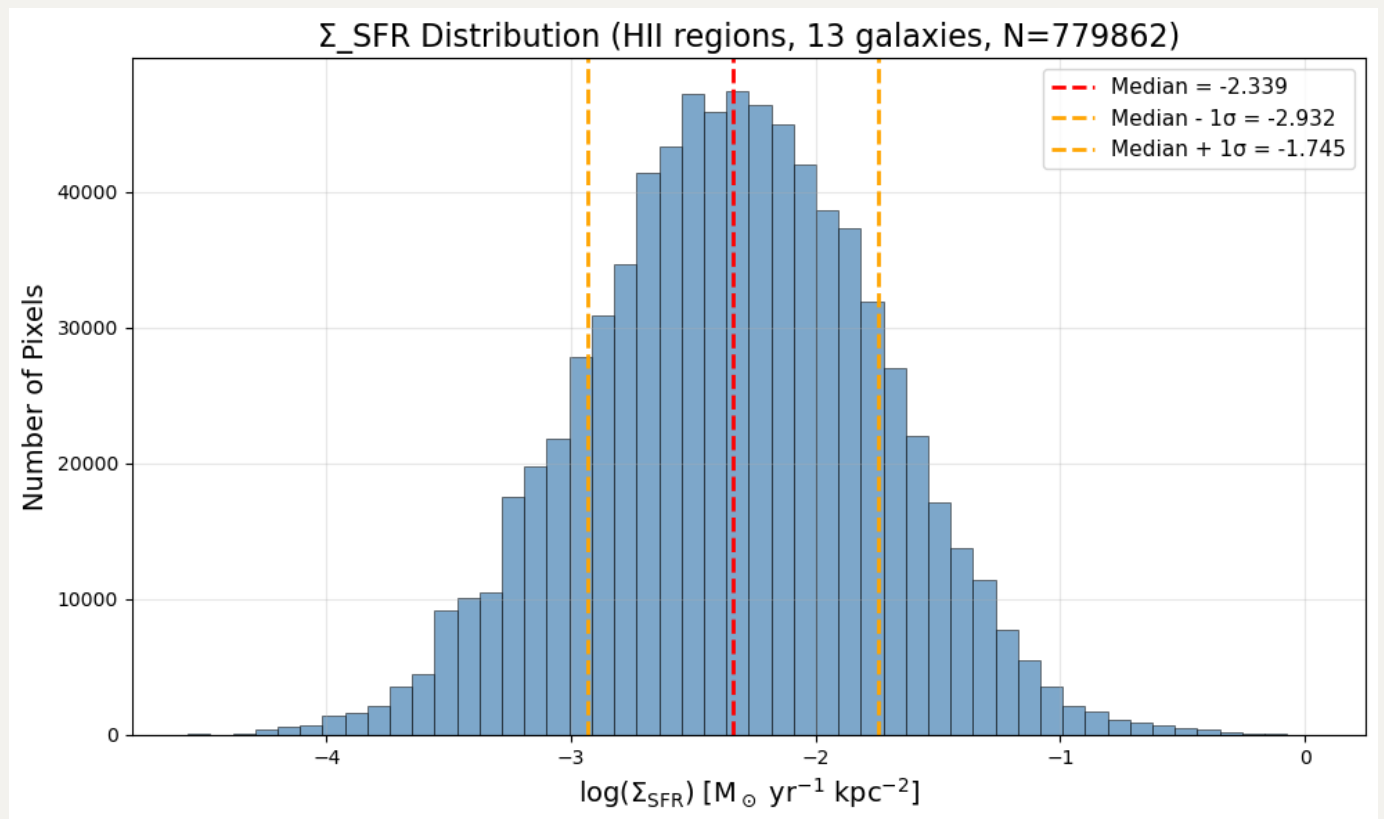


202051008 rMZR in HII But No NGC4383

Binning Effect

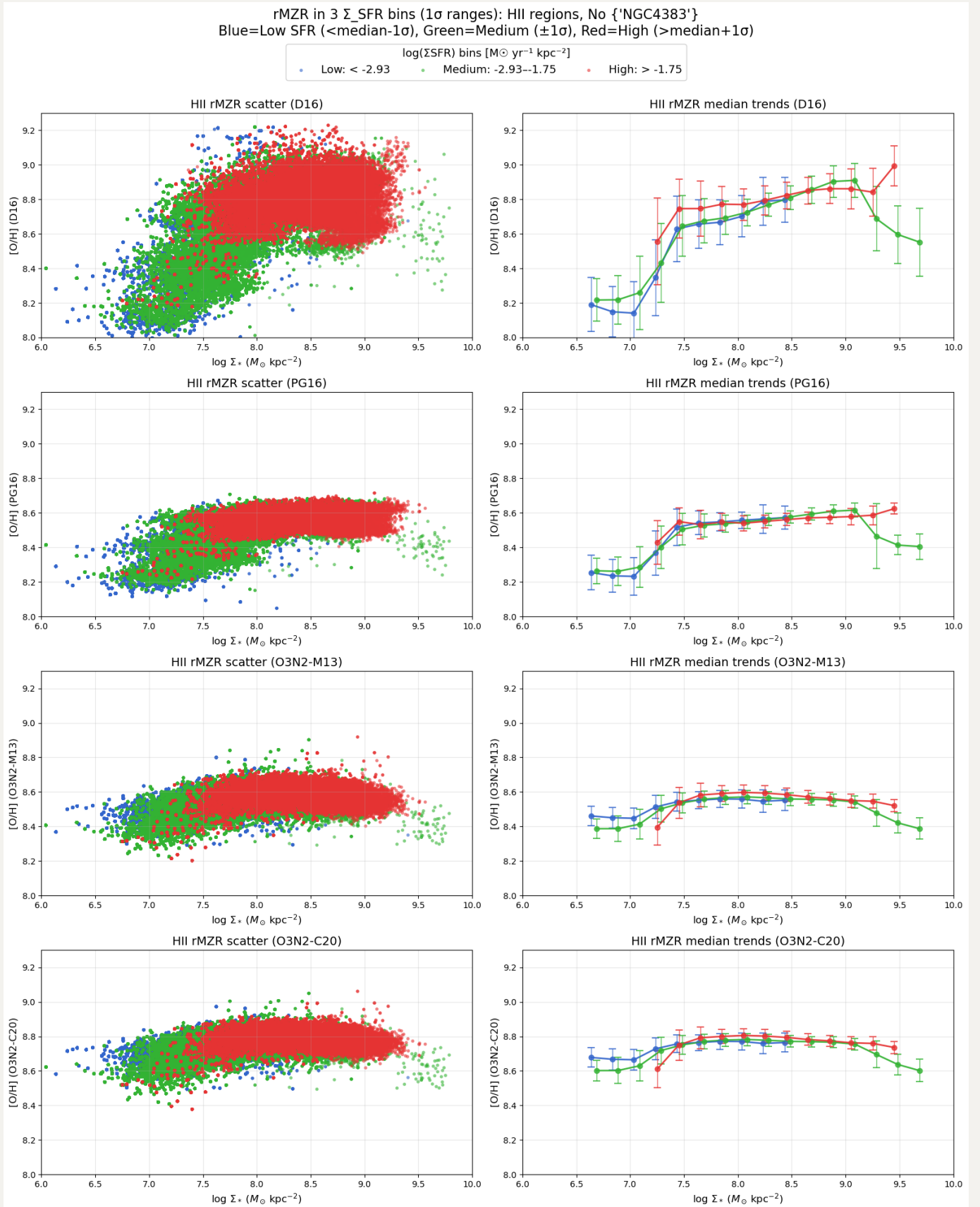
Σ_{SFR} histogram

First, check the Σ_{SFR} histogram.



Ok, so it is Gaussian shape and not surprising.

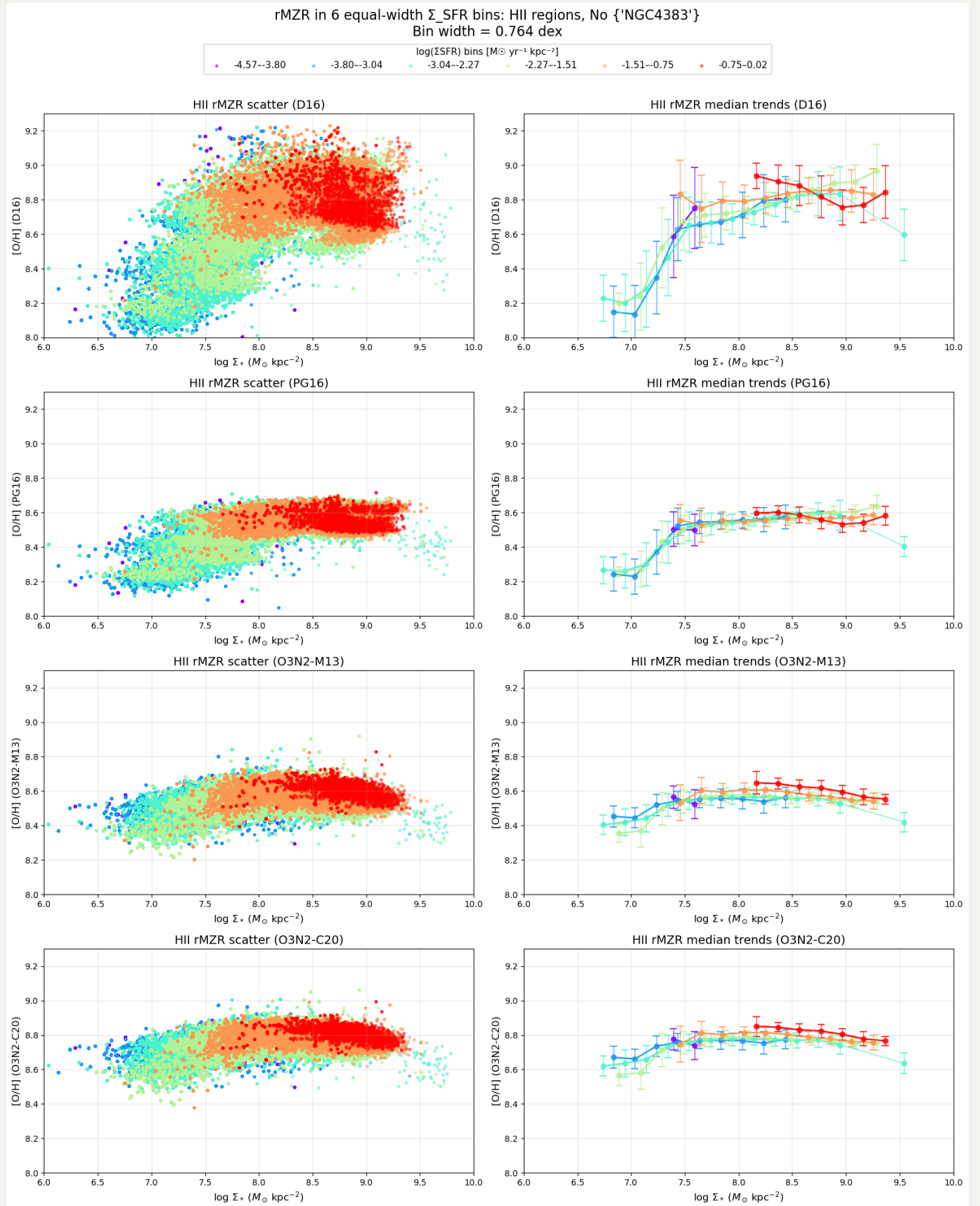
Then, we can separate into 3 Σ_{SFR} bins in this case to check the rMZR.



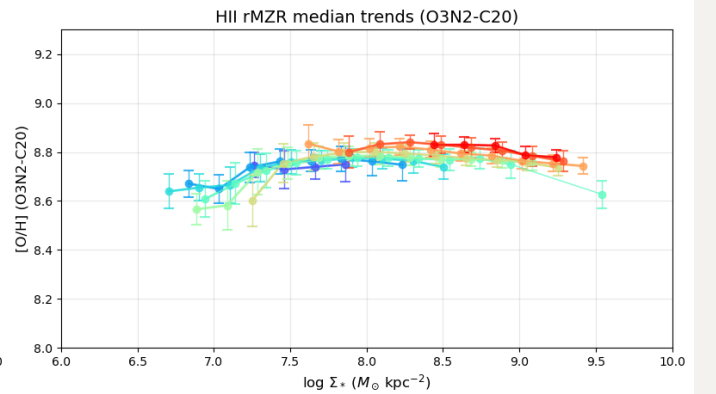
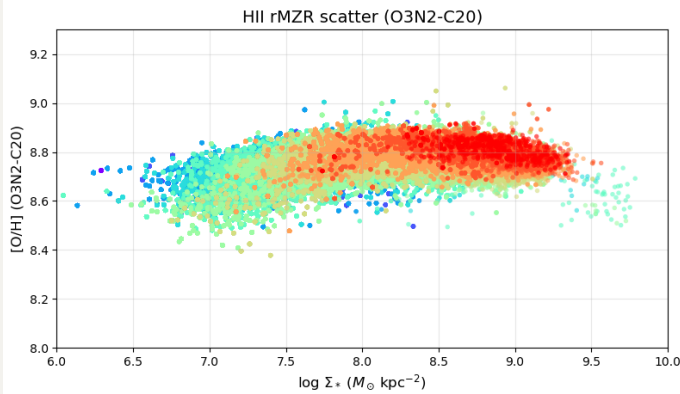
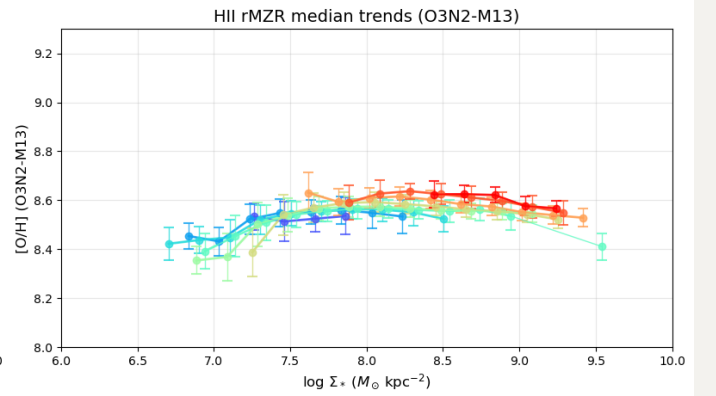
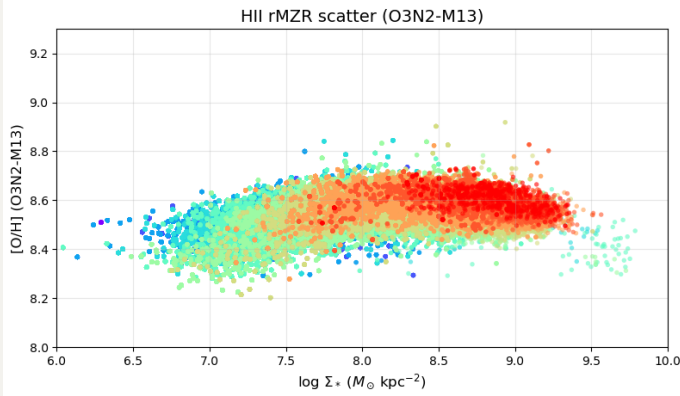
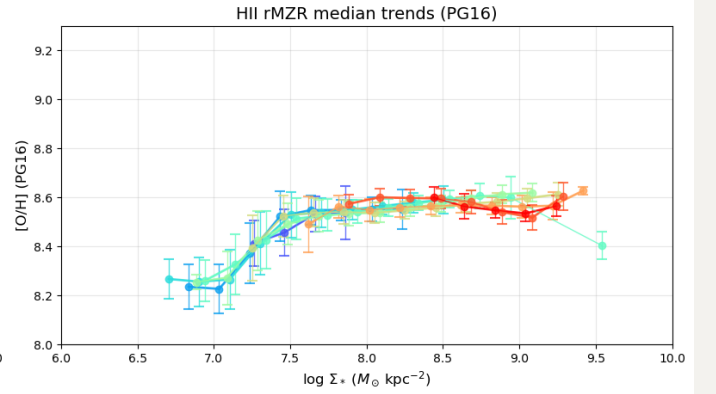
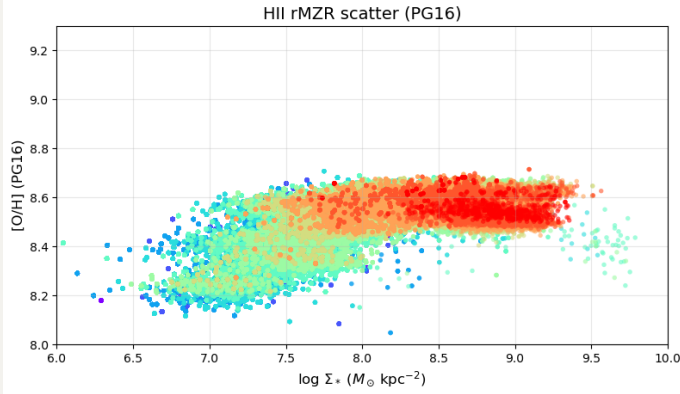
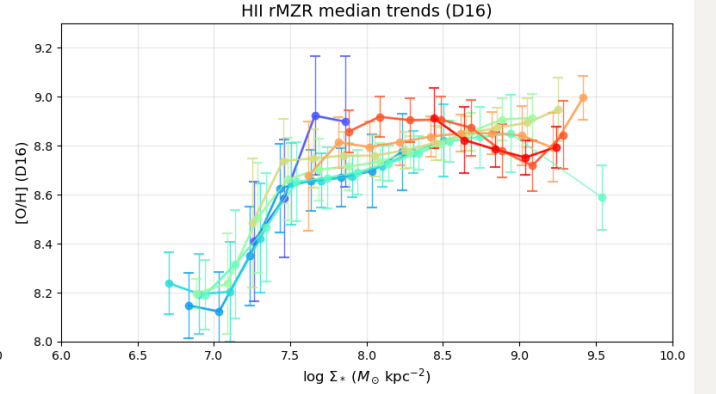
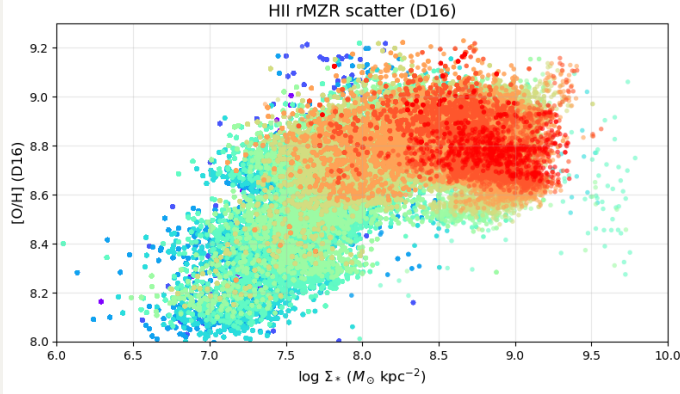
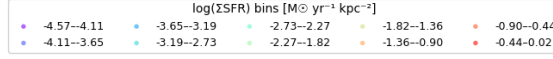
It looks like the inversion is still there.

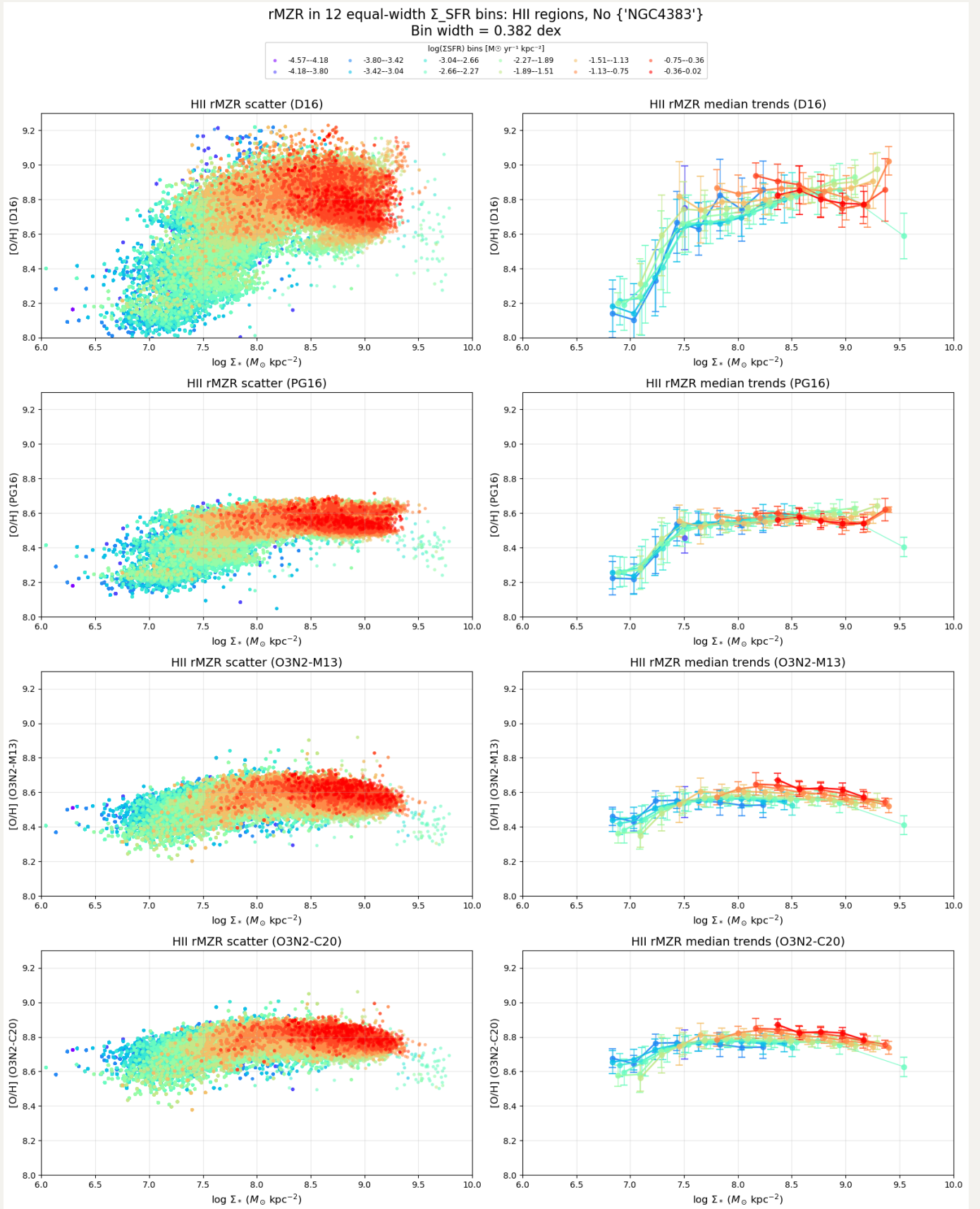
Equal binwidth, bins = 6, 10, 12

In general, all looks fine.



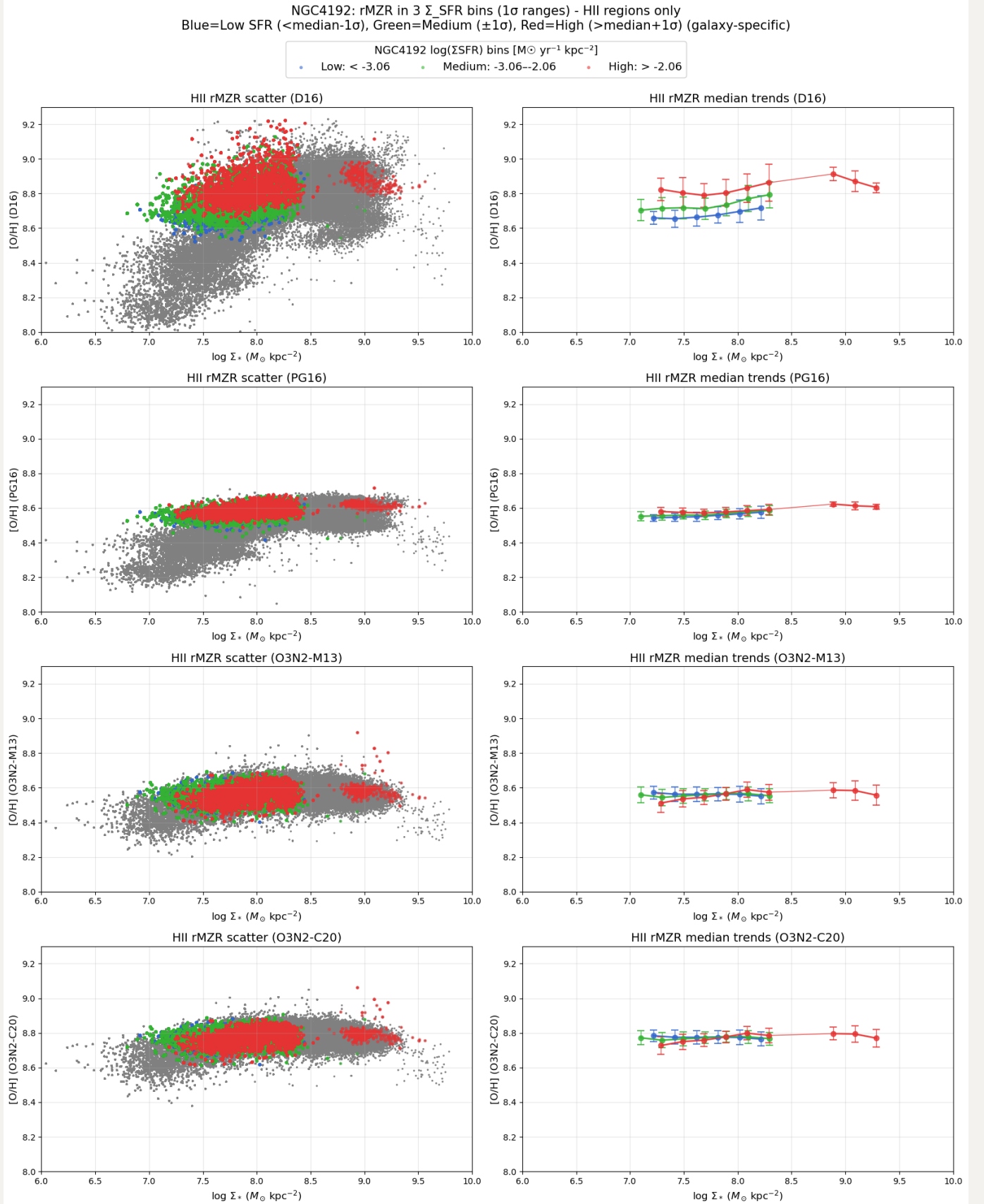
rMZR in 10 equal-width Σ_{SFR} bins: HII regions, No {'NGC4383'}
Bin width = 0.459 dex





NGC4192

However, the problem is that we can not tell if NGC4192 truly shows the inversion feature. For 3 bins, it still shows inversion; while for higher bins, the inversion looks vanish.

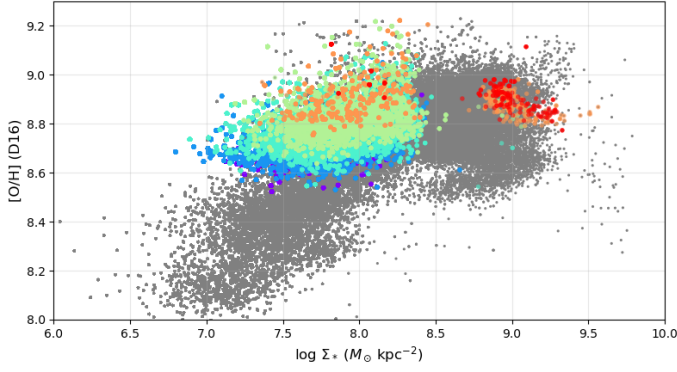


NGC4192: rMZR in 6 equal-width Σ_{SFR} bins - HII regions only
Bin width = 0.648 dex (galaxy-specific)

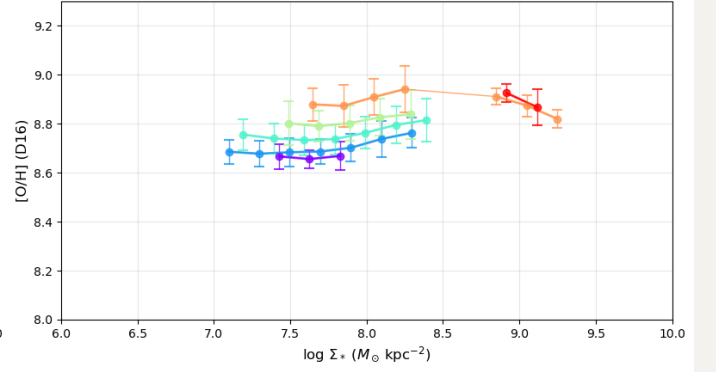
NGC4192 $\log(\Sigma_{\text{SFR}})$ bins [$M_{\odot} \text{ yr}^{-1} \text{ kpc}^{-2}$]

Bin Range	Color
-3.96--3.31	Purple
-3.31--2.66	Blue
-2.66--2.01	Cyan
-2.01--1.37	Green
-1.37--0.72	Orange
-0.72--0.07	Red

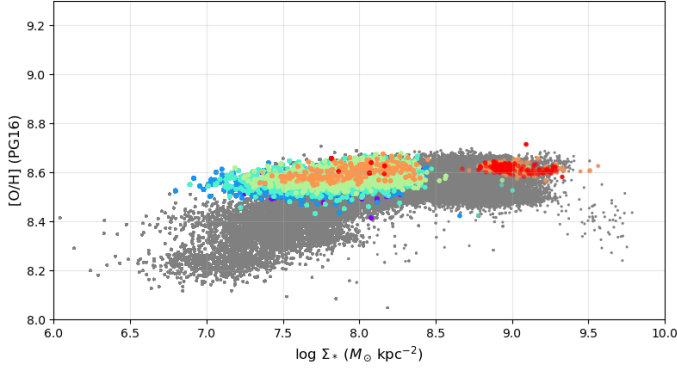
HII rMZR scatter (D16)



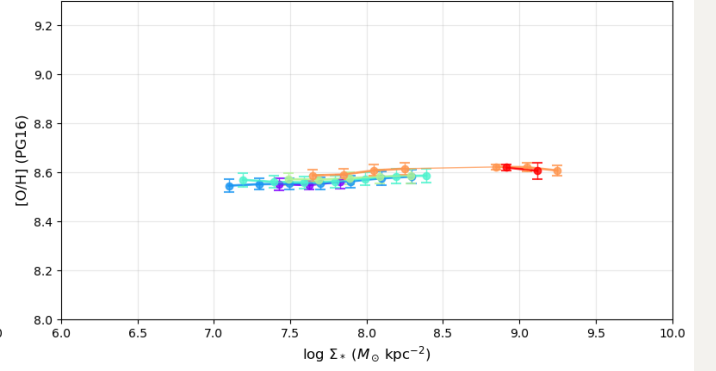
HII rMZR median trends (D16)



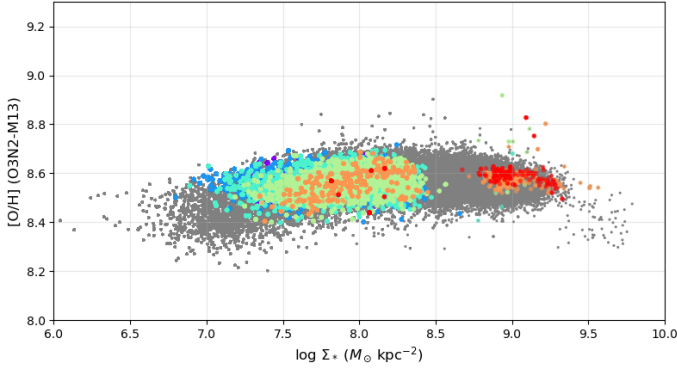
HII rMZR scatter (PG16)



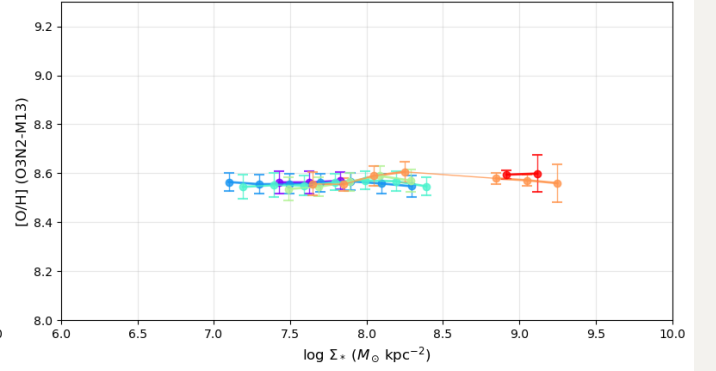
HII rMZR median trends (PG16)



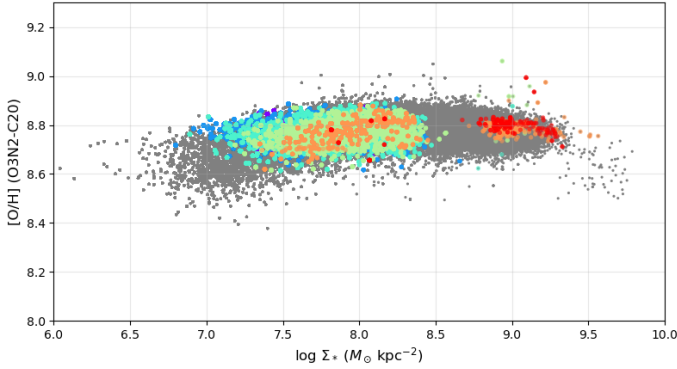
HII rMZR scatter (O3N2-M13)



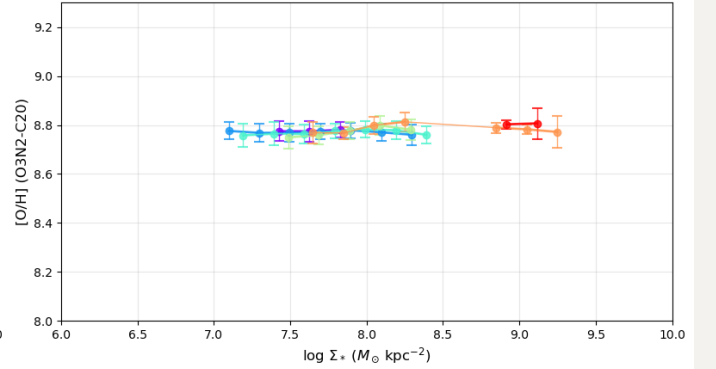
HII rMZR median trends (O3N2-M13)



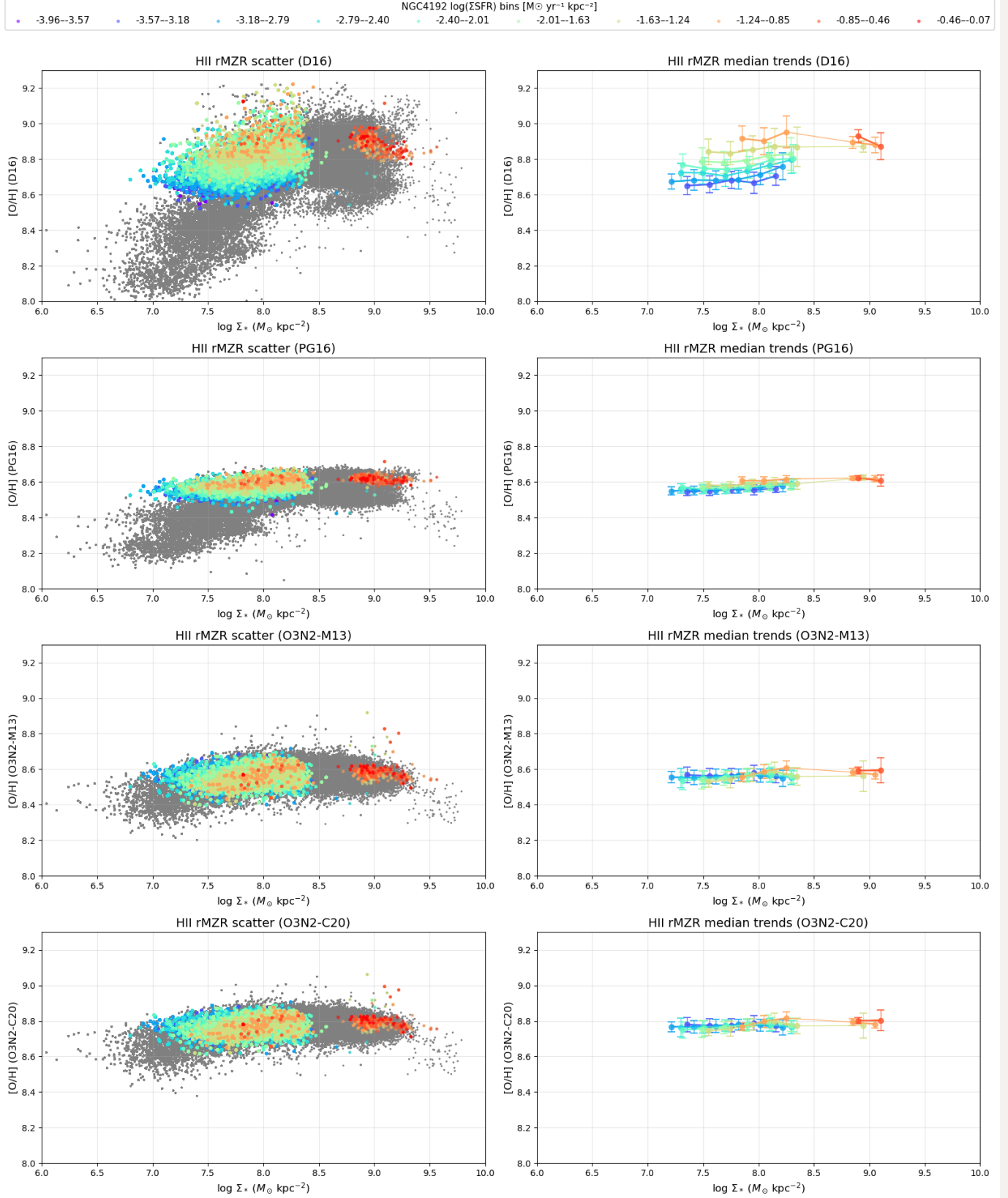
HII rMZR scatter (O3N2-C20)



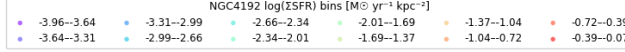
HII rMZR median trends (O3N2-C20)



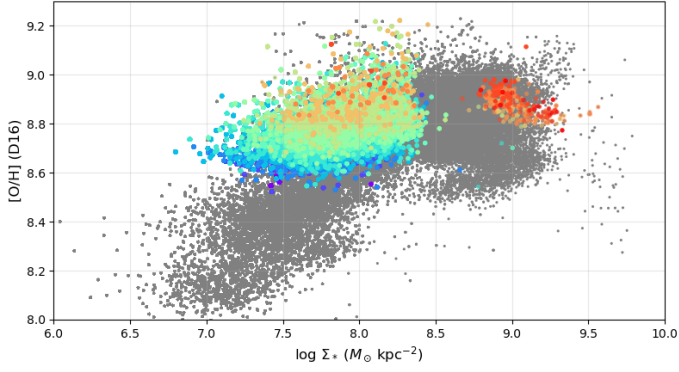
NGC4192: rMZR in 10 equal-width Σ_{SFR} bins - HII regions only
Bin width = 0.389 dex (galaxy-specific)



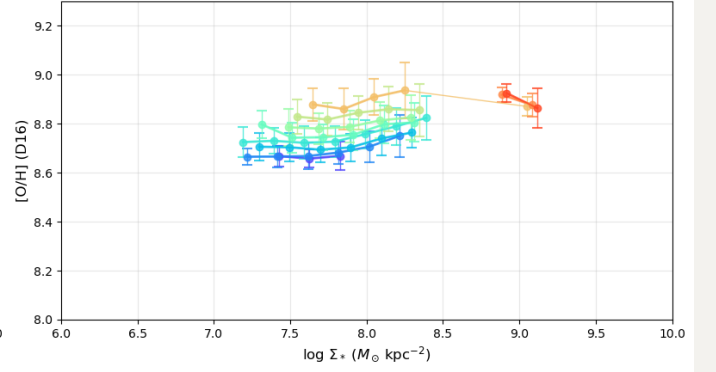
NGC4192: rMZR in 12 equal-width Σ_{SFR} bins - HII regions only
Bin width = 0.324 dex (galaxy-specific)



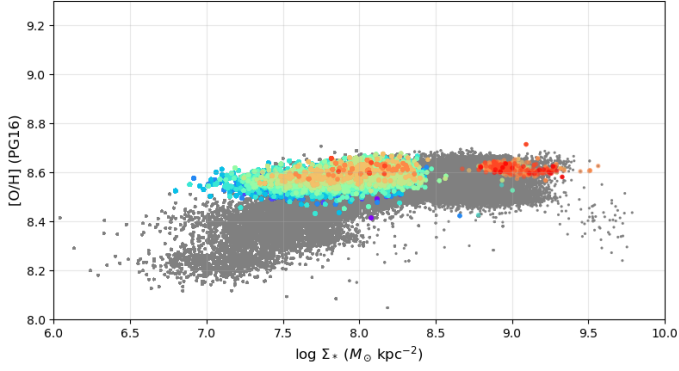
HII rMZR scatter (D16)



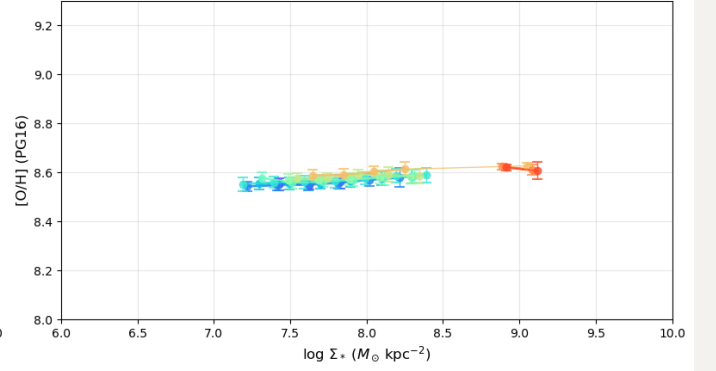
HII rMZR median trends (D16)



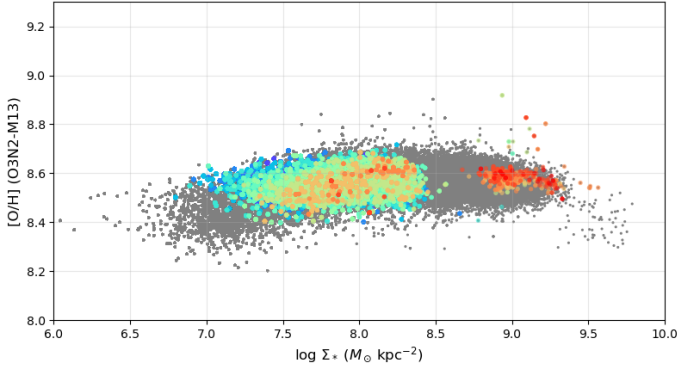
HII rMZR scatter (PG16)



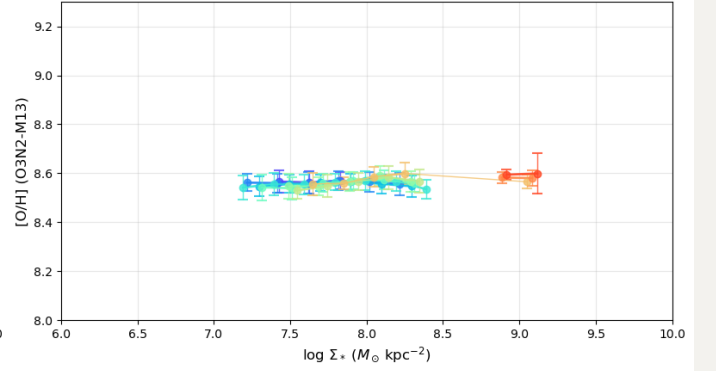
HII rMZR median trends (PG16)



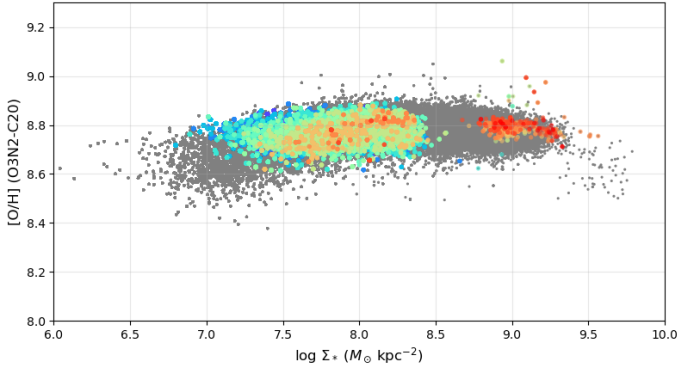
HII rMZR scatter (O3N2-M13)



HII rMZR median trends (O3N2-M13)



HII rMZR scatter (O3N2-C20)



HII rMZR median trends (O3N2-C20)

